llmXive Automated Review of DreamX-World 1.0: A General-Purpose Interactive World Model

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so errors, misreadings, and inaccuracies are likely. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is advisory, automated feedback for the authors. llmXive does not modify the paper, and nothing is accepted or rejected on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and the alignment between claims and cited evidence.

1. Discrepancy in Latency Reduction Claims (Section 3.1, Table 1) The manuscript claims in the text (Section 3.1, paragraph 2) that E-PRoPE reduces inference latency by "approximately 30%" and training time by "approximately 50%". Table 1 provides the empirical data: PRoPE latency is 80s, and E-PRoPE latency is 59s. The actual reduction is $(80 - 59)/80 = 26.25\%$. While "approximately 30%" is a common rounding convention, the specific claim of "30%" is slightly inflated compared to the provided data (26.25%). More critically, the text claims a "50%" reduction in training time, but Table 1 only reports inference latency, not training time. The claim regarding training time reduction is unsupported by the provided table. The authors should either provide the training time data in a table or adjust the text to reflect that the 50% figure is an estimate or derived from a different experiment not shown.

2. Underspecified Normalization Bounds (Section 5.1) The paper introduces a "Camera Control Metric" where the error e_{camera} is computed as the geometric mean of rotation and translation errors. The text states: "Both errors are normalized to $[0, 1]$ using empirical bounds so that higher scores indicate better adherence." However, the specific "empirical bounds" (i.e., the maximum possible error values used for normalization) are not defined in the text or the bibliography. Without these bounds, the resulting score of 73.75 is not reproducible, and the claim that the score is normalized to a specific range is unverifiable. This is a significant gap in the factual support for the evaluation results.

3. Citation Accuracy for Future-Dated Works (Section 5.3, Bibliography) The paper cites `gu2026mutualvpr` as a NeurIPS 2026 paper. The manuscript itself is dated "June 2026". Citing a paper as a "NeurIPS 2026" proceeding in a June 2026 preprint is factually problematic unless the paper has already been accepted and published in the proceedings, which is unlikely for a conference typically held later in the year. If `MutualVPR` is a preprint, the citation should be updated to reflect its arXiv status (e.g., `arXiv preprint arXiv:xxxx.xxxxx`) to accurately represent the source's current state. Claiming it as a published conference proceeding at this stage is an overstatement of its publication status.

4. Consistency of "Composable Events" Claim (Section 4, Table 2) The paper claims in Section 4 that existing systems do not provide explicit "composable events" and that DreamX-World is the first to do so. Table 2 supports this by marking other models with `\xmark` or `\pmark` for "Multi-Entity Composition" and "Inter-Object Interaction". However, the definition of "composable events" relies on the specific structured instruction format described. The claim is accurate *relative to the specific definition provided*, but the authors should ensure that the cited works (e.g., LingBot-World, HY-WorldPlay) are not being unfairly characterized if they support similar capabilities via different interfaces not captured in the table's binary checkmarks. The claim is supported by the table's internal logic, but the table's definitions are critical to the claim's validity.

Overall, the paper makes strong claims about performance and novelty, but specific numerical claims (latency reduction) and methodological details (normalization bounds)

require tighter alignment with the provided evidence or additional clarification.
Action items.

- [writing] In Section 3.1 (Camera-Aware Training), the text claims E-PRoPE reduces inference latency by 'approximately 30%' and training time by '50%'. Table 1 shows latency dropping from 80s to 59s (a 26.25% reduction). The text overstates the empirical result. Please align the claim with the table data or clarify the specific conditions under which the 30% figure was derived.
- [science] In Section 5.1 (Basic Evaluation), the text states the camera control error is computed as the geometric mean of rotation and translation errors, then normalized. However, the text does not specify the empirical bounds used for normalization to [0,1]. Without these bounds, the claim that 'higher scores indicate better adherence' and the specific score of 73.75 cannot be independently verified or reproduced.
- [writing] In Section 5.3 (Memory Evaluation), the paper claims to use 'MutualVPR' for place recognition. The bibliography lists 'MutualVPR' (gu2026mutualvpr) as a NeurIPS 2026 paper. Given the current date context (June 2026 in the paper), this is a future-dated citation. If this is a preprint, the citation should reflect its preprint status (e.g., arXiv) rather than a future conference proceeding, to ensure factual accuracy regarding the source's availability.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 is a clear and well-structured pipeline diagram that effectively visualizes the UE data generation process. The flow from scene input through exploration, validation, and rendering to output is logical, and the runtime management layer is appropriately detailed. The caption accurately summarizes the figure's content.

Figure 2

The figure illustrates a data flow but fails to visualize the core 'distillation' process described in the caption, specifically omitting the teacher model and the supervision link. Additionally, the use of emojis as component labels lacks definition, creating ambiguity regarding the model's architecture.

Figure 3

The figure displays a grid of high-quality generated images, but the caption is grammatically incomplete and the image lacks internal labels to distinguish the different scene types and camera controls described.

Figure 4

The figure effectively visualizes the three trajectory templates with clear axes and legends, but the caption requires significant cleanup to remove LaTeX artifacts and clarify the description of the translation-rotation sequence.

Figure 5

The figure effectively visualizes the human preference study with clear stacked bars and percentages. However, the caption is poorly written with missing subjects and grammatical errors that obscure the meaning of the comparison.

Figure 6

The figure illustrates the autoregressive inference pipeline but lacks a clear legend or explanation for the 'History KV cache' mechanism and the specific role of the camera pose inputs in the denoising step. Additionally, the vertical orientation of key process labels reduces readability.

Figure 7

The figure provides a clear visual overview of the system architecture, but the caption contains a missing model name and lists pipeline components that are not explicitly labeled in the diagram.

Figure 8

The figure provides a clear visual overview of the memory-conditioned training framework, but the caption fails to describe key visual components like the Error Bank and the residual-recycling path, reducing the figure's self-containment.

Figure 9

The figure effectively showcases the model's visual diversity across domains, but the caption contains a missing subject and makes a claim about control that is not visually substantiated by the static collage.
Action items.

- [science] Figure 2: The diagram depicts the 'Long-video Autoregressive Model' (student) receiving inputs from 'E-PRoPE' and 'LoRA' blocks, but the caption describes distillation from a ‘bidirectional E-PRoPE teacher’. The figure fails to visually represent the ‘teacher’ model or the specific distillation supervision signal (DMD Loss) connecting the teacher to the student, making the mechanism described in the caption invisible.

- [writing] Figure 2: The fire emoji (□) and snowflake emoji (□) are used as labels for ‘LoRA’ and ‘E-PRoPE’ components respectively, but there is no legend or caption text defining what these symbols signify (e.g., frozen vs. trainable parameters), rendering them ambiguous.
- [writing] Figure 3: The caption contains a grammatical error (‘Qualitative results of .’) where the model name is missing, and the image itself lacks any internal labels or row headers to identify the specific scene types mentioned.
- [writing] Figure 4: The caption contains raw LaTeX formatting artifacts (e.g., ‘%’, ‘$$3’, ‘blueblue’, ‘redred’) that should be cleaned for readability.
- [writing] Figure 4: The caption text for subfigure (b) lists a sequence of letters (W

  $$S$$

  L

  $$R$$

  R

  $$L$$

  L) that does not match the visual labels in the plot (S, W, L, R) or the ‘Translation-Rotation’ title.
- [writing] Figure 5: The caption contains multiple grammatical errors and missing subjects (e.g., ‘comparing with...’, ‘perspective of under...’, ‘is preferred in...’), making it unclear which model is the subject of the study.
- [writing] Figure 5: The labels ‘HY-WorldPlay’ and ‘LingBot-World’ appear on the right side of the bars, but the caption does not explicitly state that these represent the comparison baselines against the primary model.
- [science] Figure 6: The diagram depicts a ‘History KV cache’ feeding into the ‘Denoising’ block, but the caption describes ‘chunk-relative camera controls’ without explaining how the camera pose inputs (shown at the top) interact with the denoising process or the KV cache.
- [writing] Figure 6: The text labels ‘History KV cache’ and ‘Denoising’ are rotated 90 degrees, making them difficult to read and visually cluttered compared to the horizontal text.
- [writing] Figure 7: The caption reads ‘System overview of .’ with a missing model name (likely ‘DreamX-World’), which is critical for context.

- [writing] Figure 7: The caption claims the pipeline integrates ‘interaction alignment’ and ‘optimized serving’, but these specific components are not explicitly labeled in the diagram.
- [writing] Figure 8: The caption mentions ‘residual-recycling path’ and ‘perturbs conditioning tokens’, but the diagram lacks a visual representation of this path or the perturbation mechanism, making the text unverifiable from the image.
- [writing] Figure 8: The ‘Error Bank’ and ‘Save Error’ loop are shown, but the caption does not explain their role in the ‘Training framework’, creating a disconnect between the visual pipeline and the textual description.
- [writing] Figure 9: The caption contains a grammatical error where the subject is missing (‘Figure 9: generates interactive videos...’). It should read ‘DreamX-World generates...’ or similar.
- [writing] Figure 9: The caption claims ‘precise camera and event control’, but the figure is a static collage of keyframes. It does not visually demonstrate the ‘control’ aspect (e.g., via arrows, overlays, or before/after comparisons) to support this specific claim.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript exhibits a high density of domain-specific acronyms and jargon that are not defined at their first occurrence, creating a barrier for non-specialist readers. While terms like “Diffusion Transformer” (DiT) and “Variational Autoencoder” (VAE) are standard in the field, the paper frequently introduces them as acronyms (e.g., “DiT execution” in the Abstract, “VAE decoding” in Section 3.6) without spelling them out first. Similarly, method-specific acronyms such as “PRoPE” (Projective Relative Positional Encoding), “E-PRoPE” (Efficient PRoPE), “DMD” (Distribution Matching Distillation), and “RoPE” (Rotary Positional Embedding) are used extensively without explicit definition in the text body.

Specific instances requiring attention include:

- Abstract: “DiT execution”, “VAE decoding”, “E-PRoPE”, “FPS”.
- Section 2.1: “UE” (Unreal Engine), “6-DoF”, “SLERP”.
- Section 3.1: “DiT”, “PRoPE”, “RoPE”, “6-DoF”.
- Section 3.4: “DMD”, “T2V”, “I2V”.
- Section 3.5: “RL”, “DiffusionNFT”.
- Section 3.6: “KV cache”, “T2V”, “I2V”, “FPS”, “RTX”.

- Section 4.1: “VLM”, “FVD”, “FID”.

The authors should systematically scan the text to ensure every acronym is defined upon its first appearance. Additionally, terms like “WASD” and “IJKL” should be contextualized for readers unfamiliar with gaming controls. Replacing these acronyms with their full forms or providing immediate definitions will significantly improve the paper’s accessibility to a broader scientific audience.
Action items.

- [writing] Define ‘DiT’ (Diffusion Transformer) at first use in Section 3.1. Currently, the acronym appears without expansion, assuming reader familiarity with specific architecture families.
- [writing] Define ‘DMD’ (Distribution Matching Distillation) at first use in Section 3.4. The text mentions ‘DMD-style distillation’ and ‘DMD-forcing’ without explicitly stating what the acronym stands for.
- [writing] Define ‘RL’ (Reinforcement Learning) at first use in Section 3.5. While common, the paper uses ‘RL’ immediately after ‘DMD distillation’ without a prior definition in the text body.
- [writing] Define ‘VAE’ (Variational Autoencoder) at first use in Section 3.6. The term ‘VAE decoding’ appears without the full name being spelled out first.
- [writing] Define ‘FPS’ (Frames Per Second) at first use in the Abstract and Section 3.6. While standard, strict adherence to defining acronyms at first use is required for non-specialist accessibility.
- [writing] Replace ‘6-DoF’ with ‘six degrees of freedom’ at first use in Section 3.1. Acronyms for physical dimensions should be defined for general readers.
- [writing] Define ‘KV cache’ (Key-Value cache) at first use in Section 3.6. The term ‘rolling KV cache’ is used without expansion.
- [writing] Define ‘T2V’ and ‘I2V’ (Text-to-Video and Image-to-Video) at first use in Section 3.6. These abbreviations are used frequently without prior definition.
- [writing] Define ‘UE’ (Unreal Engine) at first use in Section 2.1. The text uses ‘UE-generated’ and ‘UE5’ without explicitly defining the acronym first.
- [writing] Replace ‘SLERP’ with ‘spherical linear interpolation’ at first use in Section 2.2. This is a specific mathematical operation that should be spelled out for clarity.
- [writing] Define ‘RoPE’ (Rotary Positional Embedding) at first use in Section 3.1. The text references ‘RoPE’ and ‘RoPE scaling’ without defining the acronym.
- [writing] Define ‘DiT’ again or ensure consistency if used in Section 3.2. The acronym is central to the method but needs a clear initial definition.
- [writing] Replace ‘FPS’ with ‘frames per second’ in the Abstract. The first instance of the acronym should be spelled out.

- [writing] Define 'VLM' (Vision-Language Model) at first use in Section 4.1. The text mentions 'VLM examines' without defining the term.
- [writing] Define 'FVD' and 'FID' (Fréchet Video Distance and Fréchet Inception Distance) at first use in Section 4.3. These are standard metrics but should be defined for non-specialists.
- [writing] Replace 'RTX' with 'NVIDIA RTX' or define the hardware family at first use in the Abstract. While 'RTX 5090' is specific, the brand acronym might need context for a general audience.
- [writing] Define 'PRoPE' (Projective Relative Positional Encoding) at first use in Section 3.1. The text introduces 'PRoPE' as a method name but does not explicitly state the full phrase.
- [writing] Define 'E-PRoPE' (Efficient PRoPE) at first use in the Abstract. The acronym is introduced without the full name.
- [writing] Replace 'WASD' and 'IJKL' with 'keyboard-style control signals (e.g., WASD for translation, IJKL for rotation)' in Section 2.1. While common in gaming, the specific mapping should be clear to non-gamers.
- [writing] Define 'DiT' in the Abstract. The term 'DiT execution' appears without definition.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent narrative where the proposed methods logically address the identified challenges. The progression from data curation to training stages and finally to evaluation follows a sound logical structure.

However, there are minor logical inconsistencies regarding the precise alignment between stated claims and reported data. Specifically, in the Basic Evaluation section, the text defines the 'Overall' score as the simple average of the individual metrics, yet the reported value (84.76) does not exactly match the arithmetic mean of the provided numbers (which calculates to ~84.77). While likely a rounding artifact, the explicit definition of the calculation method makes this a minor logical gap.

Furthermore, the claim that E-PRoPE reduces training time by 'approximately 50%' is stated as a fact but lacks the supporting data point in the provided tables or text, unlike the inference latency claim which is backed by Table 1. This leaves the premise of the 50% training reduction unsupported by the evidence presented in the manuscript.

Finally, the causal link between the problem of 'reduced motion smoothness' in chunk-wise inference and the solution of 'repeating long-video DMD training' is asserted without a clear mechanistic explanation. The text does not explain *why* repeating the distillation step specifically mitigates the smoothness issue at chunk boundaries, creating a small gap in the logical flow of the argument. These issues are fixable by clarifying the calculation, providing the missing training time data, or elaborating on the mechanism of the repeated DMD step.

Action items.

- [writing] Table 1 claims 'Overall' is the average of metrics, but the mean of listed values (~84.77) differs from the reported 84.76, creating a minor inconsistency between the stated formula and result.
- [writing] Section 3.1 claims E-PRoPE reduces training time by ~50% but provides no data to support this, unlike the inference latency claim which is backed by Table 1.
- [writing] Section 3.4 asserts repeating DMD training fixes chunk-boundary smoothness issues but fails to explain the causal mechanism linking the solution to the specific problem.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper exhibits significant overreach in its characterization of the model's generalization capabilities and the immediacy of its deployment metrics.

First, the central claim of being a "General-Purpose" world model (Title, Abstract, Introduction) is not fully supported by the evaluation scope. While the authors combine data from Unreal Engine, real-world videos, and specific game datasets, the evaluation benchmarks (Omni-WorldBench) and qualitative results appear to focus on standard navigation and camera control. There is no evidence provided that the model can handle arbitrary, unseen stylized domains (e.g., abstract art, specific anime styles) or complex physical interactions outside the training distribution without explicit fine-tuning. The assertion that a single model operates seamlessly across "photorealistic, game-style, and stylized domains" implies a level of domain-agnostic robustness that the current data mix and evaluation do not substantiate. The paper should temper this claim to reflect the specific domains covered by the training data.

Second, the performance claim of "up to 16 FPS on eight RTX 5090 GPUs" (Abstract, Introduction, Inference Acceleration) is scientifically premature. As of the paper's date (June 2026), the RTX 5090 is not a commercially available or benchmarked hardware standard. Presenting specific FPS numbers on unreleased hardware as a factual result rather than a theoretical projection or a benchmark on equivalent current-generation hardware (e.g., RTX 4090) constitutes an overstatement of the system's current real-world readiness. This risks misleading readers regarding the actual latency and hardware requirements for deployment.

Finally, the claim of "outperforming" baselines in "overall score" (Table 1, Table 2) relies heavily on the authors' own proposed benchmark (Omni-WorldBench), where several authors of this paper are also listed as contributors to the benchmark paper (Wu et al., 2026). While the metrics are detailed, the paper does not sufficiently address the potential for bias in the metric design or the weighting scheme that leads to the "overall score." The conclusion that the model is superior is based on a specific, self-defined aggregate that may not capture all aspects of world model quality (e.g., long-term physical consistency, complex event reasoning) where baselines might perform better. The paper should avoid definitive "outperformance" language without a more robust, independent, or multi-benchmark validation.

Action items.

- [science] The claim of 'general-purpose' operation across photorealistic, game, and stylized domains is over-extended. The evaluation data sources (SpatialVID, RealEstate10K, Sekai) are heavily skewed towards real-world and specific game engines. The paper lacks evidence that the model generalizes to unseen stylized domains (e.g., anime, oil painting) without specific fine-tuning, yet the abstract and introduction assert broad domain agnosticism.
- [writing] The '16 FPS on eight RTX 5090 GPUs' claim is premature and potentially misleading. The RTX 5090 is a hypothetical/unreleased hardware at the time of this preprint (June 2026). Presenting performance benchmarks on non-existent hardware as a concrete result overstates the current state of the system's deployability. This should be qualified as a projection or based on equivalent current-gen hardware.
- [science] The claim that the model 'outperforms' HY-WorldPlay 1.5 and LingBot-World in 'overall score' relies on a custom, non-standardized metric suite (Omni-WorldBench) where the authors are also listed as contributors (see bib entry wu2026omniworldbench). The paper does not sufficiently justify why this specific weighted average of metrics constitutes a definitive 'outperformance' over baselines that may excel in unmeasured dimensions (e.g., physics simulation, specific interaction types).

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript presents a sophisticated interactive world model with significant capabilities in camera control, long-horizon generation, and event manipulation. From a safety and ethics perspective, the primary concerns revolve around data provenance, potential dual-use applications, and the transparency of evaluation protocols.

First, regarding data privacy and consent, Section 3.1.2 ("Real-world and Game Data") lists the ingestion of real-world video datasets (SpatialVID, RealEstate10K, Sekai, DL3DV) for training. While the paper details the technical pipeline for camera pose recovery and filtering, it omits any discussion of the ethical acquisition of this data. Specifically, there is no mention of whether the original datasets contained personally identifiable information (PII), how such data was handled (e.g., blurring faces, removing license plates), or confirmation that the usage complies with the specific licenses and terms of service of the source datasets. Given the increasing scrutiny on training data for generative models, a dedicated statement or subsection addressing data privacy, consent, and license compliance is required to ensure the research adheres to ethical standards.

Second, the paper highlights dual-use risks associated with the model's advanced control features. The "Event Instruction Tuning" (Section 4.3) and "Composable Events" capabilities allow users to specify complex interactions, such as object collisions, handoffs, and multi-entity dynamics. While framed as a feature for realistic simulation, these capabilities could be misused to generate highly convincing disinformation, deepfakes of specific individuals in compromising situations, or realistic depictions of harmful events. The manuscript currently lacks a discussion on the potential for misuse, the implementation of safety guardrails (e.g., content filters, refusal mechanisms for harmful prompts), or the authors' strategy for responsible release. A discussion on these risks and the measures taken to mitigate them is essential for a complete safety assessment.

Finally, the evaluation methodology raises minor ethical questions. The "Artifact Detection Metric" (Section 6.1) utilizes a proprietary VLM (Gemini-3.1-Pro) to judge generated content, and the "Human Preference Study" (Section 6.4) involves human assessors. The paper does not specify the ethical guidelines provided to human assessors, their demographic diversity, or how potential biases were mitigated. Furthermore, there is no mention of whether the human study was approved by an Institutional Review Board (IRB) or if informed consent was obtained from participants. While the study appears to be a standard preference test, explicit confirmation of ethical oversight and the protocols used to protect participants would strengthen the paper's ethical standing.

In summary, while the technical contributions are significant, the paper requires revisions to explicitly address data privacy, dual-use risks, and the ethical oversight of its evaluation procedures.

Action items.

- [writing] The paper describes training on real-world videos (SpatialVID, RealEstate10K, Sekai, DL3DV) with recovered camera poses but lacks explicit statements regarding data privacy, consent, or compliance with the original dataset licenses. A dedicated 'Data Ethics' or 'Privacy' subsection is required to confirm that all real-world data usage adheres to the source terms and that no personally identifiable information (PII) was retained or exposed.

- [writing] The 'Event Instruction Tuning' and 'Composable Events' capabilities allow users to generate specific interactions (e.g., collisions, handoffs) and modify world states. The paper does not address potential dual-use risks where these features could be exploited to generate realistic disinformation, deepfakes of specific individuals, or harmful scenarios. A discussion on safety guardrails, content filtering, or responsible release policies is necessary.
- [writing] The evaluation section relies on a VLM (Gemini-3.1-Pro) for artifact detection and human preference studies. The paper does not disclose the demographic composition of the human assessors or the specific safety guidelines provided to them to prevent bias or the generation of harmful content during the study. Clarification on the ethical oversight of the human evaluation process is needed.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a comprehensive system for interactive world modeling, but the scientific evidence supporting the quantitative claims requires strengthening to meet publication standards.

Statistical Rigor and Variance: The primary limitation in the evidence provided is the absence of statistical variance metrics. In Tables 1 (Basic Evaluation), 2 (Long-Horizon), and 3 (Memory Consistency), the authors report single scalar values for all models (e.g., DreamX-World 1.0: 84.76 vs. HY-WorldPlay 1.5: 80.79). There is no indication of whether these results are averages over multiple random seeds, different test splits, or a single run. In generative modeling, performance can vary significantly based on initialization and stochastic sampling. Without standard deviations or confidence intervals, it is impossible to determine if the reported improvements are statistically significant or within the noise floor of the evaluation pipeline. The authors must re-run evaluations with multiple seeds (e.g., n=5) and report mean ± std.

Evaluation Metric Validity: The "Artifact Detection Metric" (Section 5.1) relies entirely on a single VLM (Gemini-3.1-Pro) to judge binary pass/fail outcomes for defects like "duplicated limbs" or "geometric pass-through." The paper does not provide evidence that this specific VLM aligns with human perception for these specific artifacts. A robust evaluation would include a human-in-the-loop validation study (e.g., calculating Cohen's Kappa between the VLM and human annotators on a subset of 100-200 samples) to establish the metric's validity. Relying on a black-box VLM without calibration risks measuring the VLM's biases rather than actual model artifacts.

Memory Consistency Significance: In Section 5.3, the authors introduce a "gain-based" scoring system for memory consistency. While the methodology of comparing revisit pairs to non-revisit baselines is sound, the results in Table 3 lack context regarding sample size. How many revisit pairs were detected across the test set? A gain of 0.232 in DINO-Sim might be highly significant with 10,000 pairs but negligible with 50. Furthermore, no statistical tests (e.g., t-tests or Mann-Whitney U tests) are reported to confirm that the gains are distinguishable from zero or from the baselines.

Reproducibility of Baselines: The comparison against HY-WorldPlay 1.5 and LingBot-World assumes these baselines were evaluated under identical conditions (resolution, frame rate, prompt distribution). The paper mentions augmenting the trajectory set for the camera control metric but does not explicitly state whether the baselines were re-evaluated on this *augmented* set or if the authors used the original benchmark scores. If the baselines were not re-evaluated on the exact same augmented test set, the comparison is invalid. The authors must clarify the evaluation protocol for all models to ensure a fair "apples-to-apples" comparison.

Action items.

- [science] The evaluation section (Section 5) lacks statistical rigor. Tables 1, 2, and 3 report single-point scores without standard deviations, confidence intervals, or p-values. Given the claim of outperforming baselines, the authors must report variance across multiple seeds or test splits to rule out random fluctuation.
- [science] The 'Artifact Detection Metric' relies on a single VLM (Gemini-3.1-Pro) without reporting inter-rater reliability or calibration against human judgment. The binary pass/fail judgment is subjective; the authors should provide a validation study showing the VLM's agreement with human annotators on a held-out set.
- [science] The memory consistency evaluation (Table 3) reports 'gain-based' scores but does not specify the sample size (number of revisit pairs) or the statistical test used to determine if the observed gains are significant. Without this, the claim of 'stronger memory' is not statistically supported.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in the evaluation section requires clarification to ensure reproducibility and validity of the reported claims.

First, regarding the Camera Control Metric (Section 5.1, Eq. 1), the paper defines the error as the geometric mean of rotation and translation errors ($e_{\text{camera}} = \sqrt{e_\theta \cdot e_t}$). However, the text states these are "normalized to yield a final score" without defining the normalization function, the empirical bounds used for scaling, or the specific mapping to the [0, 100] range. Since the final scores in Table 1 (e.g., 73.75) are the primary evidence for the model's superiority, the exact transformation from raw geometric error to the reported score must be explicitly defined to allow independent verification.

Second, the Memory Evaluation (Section 5.3) introduces a novel "Gain-based Scoring" protocol. While the methodology of subtracting baseline scores is sound, the results in Table 3 are presented as point estimates (e.g., ΔPSNR = 3.92) without any measure of variance (standard deviation, standard error) or confidence intervals. In generative modeling, where results can vary significantly across seeds or random initializations, reporting only the mean gain is insufficient. The authors should report the standard deviation across multiple runs or provide confidence intervals to demonstrate that the observed gains are statistically significant and not due to random fluctuation.

Third, the Human Preference Study (Section 5.4) reports win/tie/loss rates (e.g., 57.5% win rate) but omits the sample size (N) of the study. Without knowing the total number of comparisons or the number of human annotators, it is impossible to assess the statistical power of the study or calculate confidence intervals for the preference rates. A simple binomial proportion confidence interval (e.g., Wilson score interval) should be reported to contextualize the significance of the "win" rates against the baselines.

Finally, while the paper compares against baselines, it does not explicitly state whether the comparisons in Tables 1 and 2 are statistically significant. Given the small margins in some metrics (e.g., Overall score 84.76 vs 80.79), a formal statistical test (such as a paired t-test or Wilcoxon signed-rank test) across multiple generated samples would strengthen the claim of superiority.

Action items.

- [science] The camera control error formula (Eq. 1) defines a geometric mean of rotation and translation errors, but the text fails to specify the normalization bounds or the exact mapping function used to convert these errors into the [0,100] score reported in Table 1. Without this, the metric is not reproducible.
- [science] The 'Gain-based Scoring' for memory evaluation (Section 5.3) reports differences against a baseline but omits the standard deviation or confidence intervals for these gains. Given the high variance typical in generative model evaluation, statistical significance testing (e.g., paired t-tests) is required to validate the claimed improvements.
- [science] The human preference study (Section 5.4) reports win/tie/loss percentages but does not state the total number of trials (N) or the number of human assessors. This prevents the calculation of confidence intervals or the assessment of statistical power for the reported preferences.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high overall standard of academic writing, with clear structure, logical progression, and generally precise terminology. The abstract and introduction effectively frame the problem and contributions. However, several minor grammatical errors, phrasing inconsistencies, and instances of redundancy detract from the polish of the text.

Specifically, in Section 3.1, the transition between the theoretical justification for downsampling tokens and the practical implementation of E-PRoPE feels slightly abrupt. The argument that "view-dependent high-level semantics" do not require full-resolution tokens would benefit from a more explicit explanation of why high-level semantics are preserved at lower resolutions in this specific context.

In Section 3.2, the phrase "In specific" is non-standard; "Specifically" is the preferred adverbial form. Similarly, in Section 3.4, the omission of the article "a" in "train few-step autoregressive model" is a clear grammatical oversight. These errors, while not obscuring meaning, suggest a need for a final proofread.

Section 4.1 contains a repetitive sentence structure in the description of the artifact detection metric, where the definition of the metric's focus is restated immediately after being introduced. Streamlining this would improve the flow. Additionally, minor inconsistencies in citation spacing (e.g., missing space before `\citep`) appear in the Related Work section, which, while often a LaTeX style issue, can affect the visual readability of the compiled text.

Overall, the writing is strong and the paper is highly readable, but addressing these specific grammatical and stylistic points will elevate the manuscript to a polished state suitable for publication.

Action items.

- [writing] In Section 3.1 (Camera-Aware Training), the sentence 'We argue that PRoPE primarily captures the view-dependent high-level semantics' lacks a clear logical bridge to the subsequent claim that full-resolution tokens are unnecessary. Clarify the causal link between semantic abstraction and token downsampling to improve argumentative flow.
- [writing] In Section 3.2 (Memory-Conditioned Scene Persistence), the phrase 'In specific, we use camera pose...' contains a grammatical error. It should be corrected to 'Specifically, we use...' or 'In particular, we use...' for standard academic phrasing.
- [writing] In Section 3.4 (Autoregressive Long Video Generation), the sentence 'We train few-step autoregressive model using causal forcing...' is missing the indefinite article 'a' before 'autoregressive model'. Correct to 'a few-step autoregressive model'.
- [writing] In Section 4.1 (Basic Evaluation), the paragraph describing the 'Artifact Detection Metric' repeats the phrase 'critical defects and failures during the generation process' almost verbatim from the previous sentence. Rephrase to avoid redundancy and improve conciseness.

- [writing] In Section 5 (Related Work), the paragraph on 'World Model Evaluation' contains a citation formatting inconsistency: 'Omni-WorldBench\citep{wu2026omniworldbench}' lacks a space before the citation command, while others have it. Ensure consistent spacing for readability.